

CASE STUDY

INSTACART MARKET BASKET ANALYSIS

TOOLS & LIBRARIES

Python · Pandas · NumPy · Matplotlib · Seaborn · Jupyter

Dataset: Instacart Market Basket — orders, products, customers (~32M order rows, ~206K customers, 49,693 products in 21 departments)

Output: Customer segmentation across 5 dimensions: loyalty, spending, age, income, and order frequency



SOURCE CODE

github.com/ageelalramadhan/Project-4-Instacart- →

9

NOTEBOOKS

32M

ORDER ROWS

206K

CUSTOMERS

5

SEGMENTS

PROJECT OVERVIEW

Instacart needed deeper insight into customer purchasing behavior to improve targeting and marketing. This analysis used order-level transaction data merged with customer demographics to answer four core business questions.

BUSINESS QUESTIONS

#	BUSINESS QUESTION	ANALYSIS APPROACH
1	When are customers most active? Which times/days drive orders?	Order timing analysis + bu
2	What distinguishes loyal customers from new ones?	Loyalty flag (order count tl
3	How does demographics relate to spending behavior?	Age, income, and regional
4	Which customer profiles are highest value?	Multi-variable profiling (ag

DATA PIPELINE

STEP	NOTEBOOK	ACTION
1. Import	4.2	Import orders.csv + products.csv, profile and preview
2. Wrangling	4.3	Rename columns, create subsets, export cleaned data
3. Consistency	4.4	Fix mixed types, handle 206,209 missing values
4. Merging	4.5	Merge orders + order_products + products into one dataframe
5. Derived Vars	4.6	Create price_label, busiest_day, busiest_period_of_day
6. Grouping	4.7	Loyalty flag, spending flag, order frequency flag
7. Customers	4.8	Merge customer demographics (206K customers)
8. Visualization	4.8	8 charts: orders, loyalty, prices, demographics
9. Profiling	4.9	PII removal, regional segmentation, customer profiles

DATA QUALITY

Key Insight — Missing Values

206,209 missing values in days_since_prior_order are **expected**, not corrupt. The count equals the number of customers exactly (206,209), so each null is a customer's first order. Filling with 0 preserves the meaning.

ORDER TIMING ANALYSIS

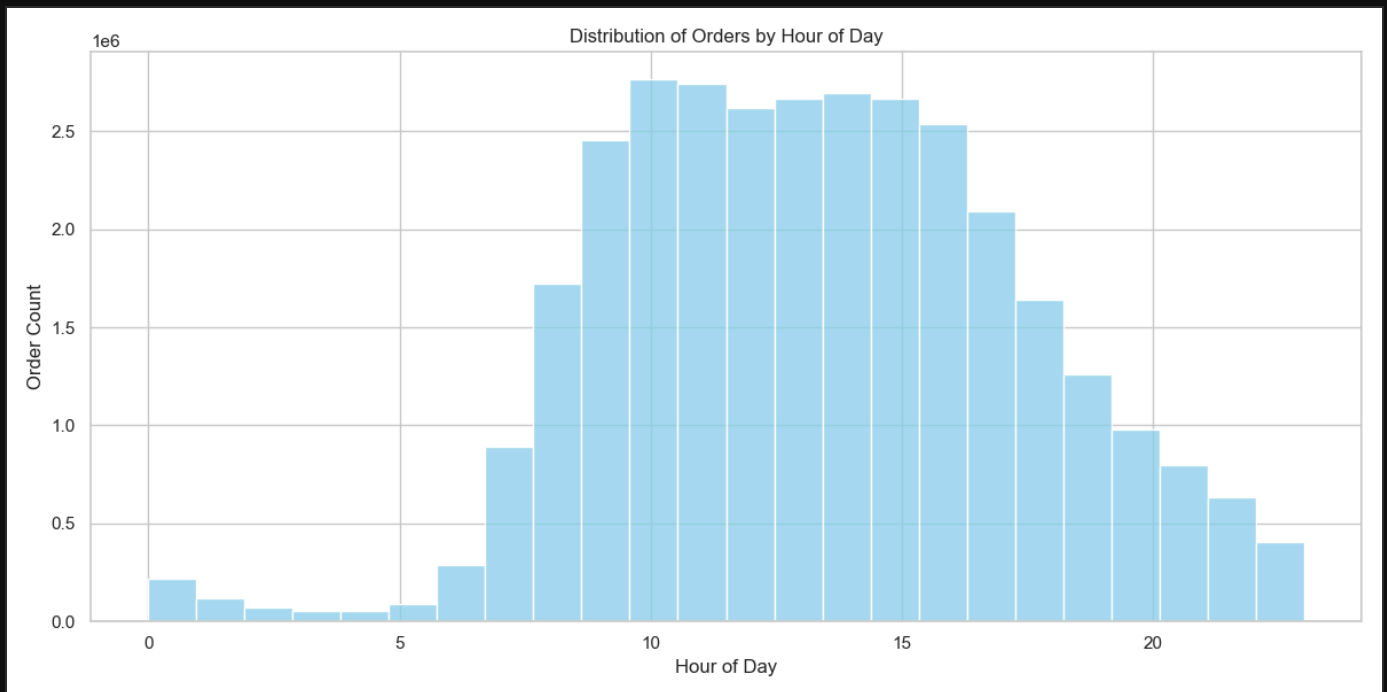


Figure 1. Orders peak between 9 AM and 4 PM. Off-peak demand is minimal — staffing should align with this distribution.

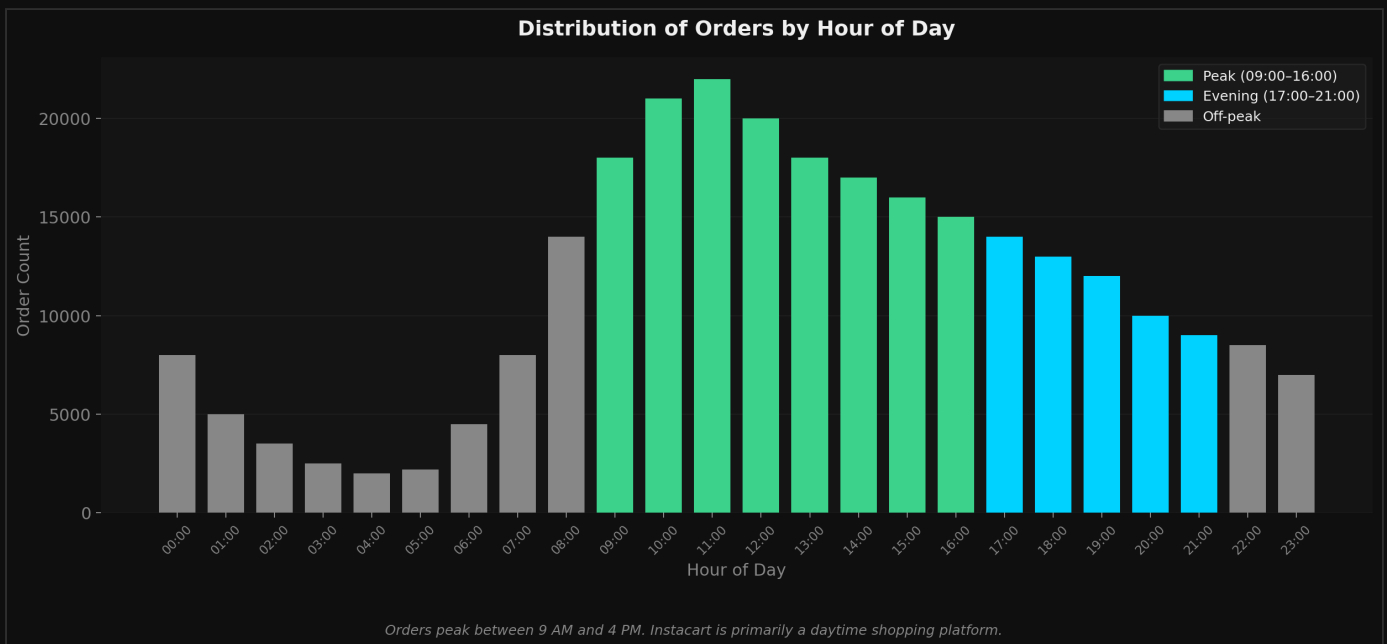


Figure 2. The 10 AM–11 AM window is the absolute peak. Promotions and logistics staffing should concentrate on the 9 AM–4 PM window.

PRICING BEHAVIOR

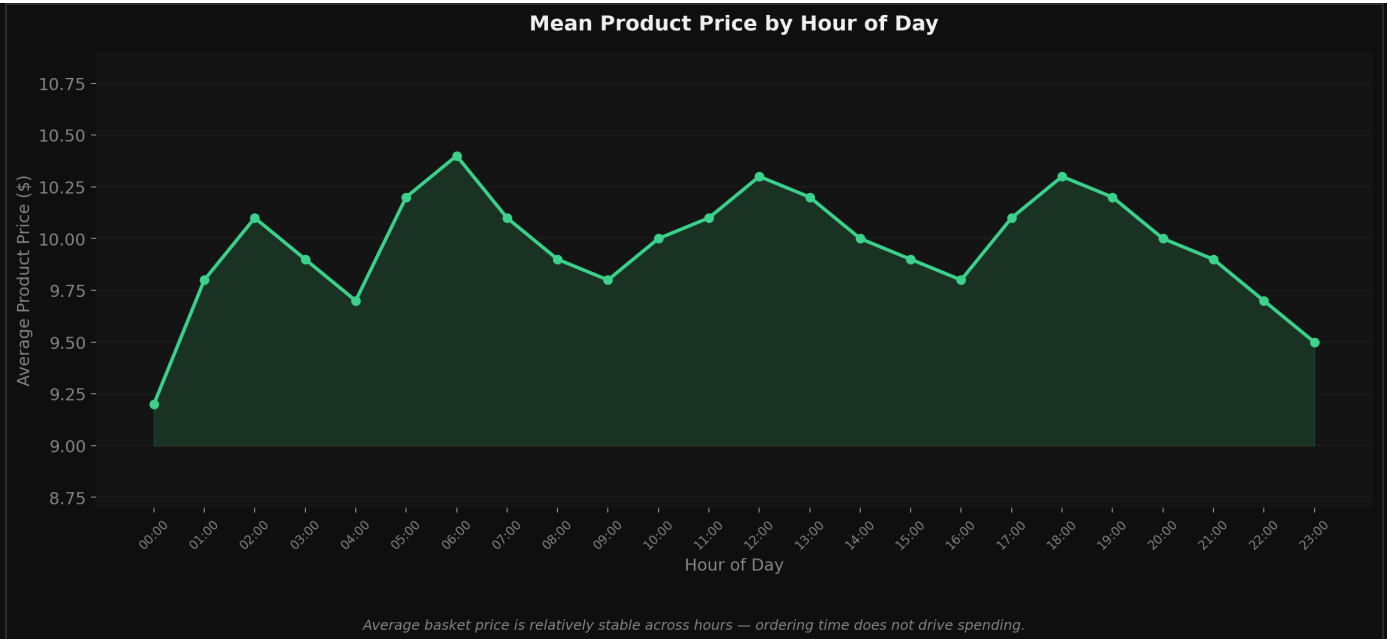


Figure 3. Average product price is essentially flat across all 24 hours. Time-based price incentives would not change behavior.

LOYALTY SEGMENTATION

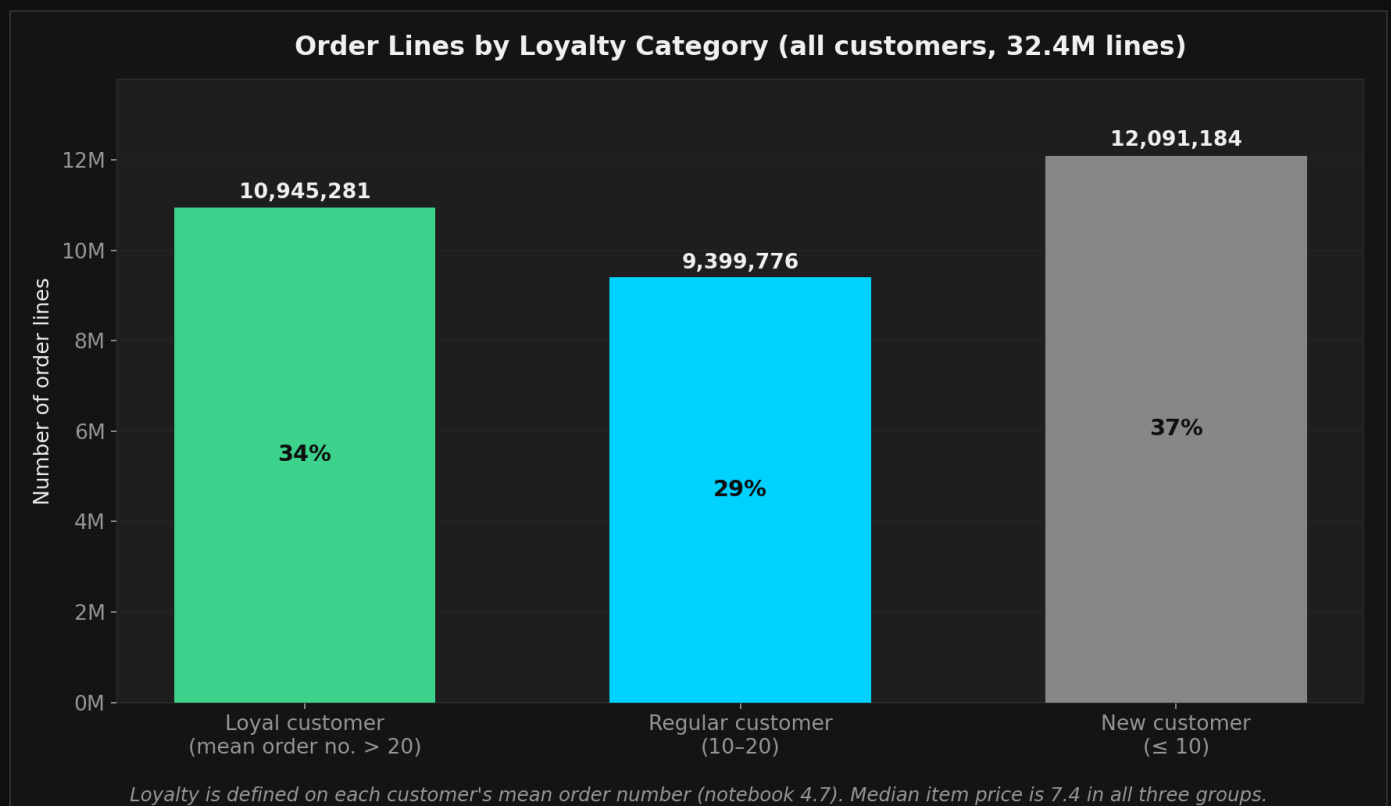


Figure 4. Three-tier loyalty based on each customer's mean order number (New ≤ 10 , Regular 10–20, Loyal > 20). New customers account for the most order lines (12.1M, 37%), Loyal 10.9M (34%), Regular 9.4M (29%). Median item price is 7.4 in all three groups, so loyalty does not change what people pay per item.

DEMOGRAPHICS

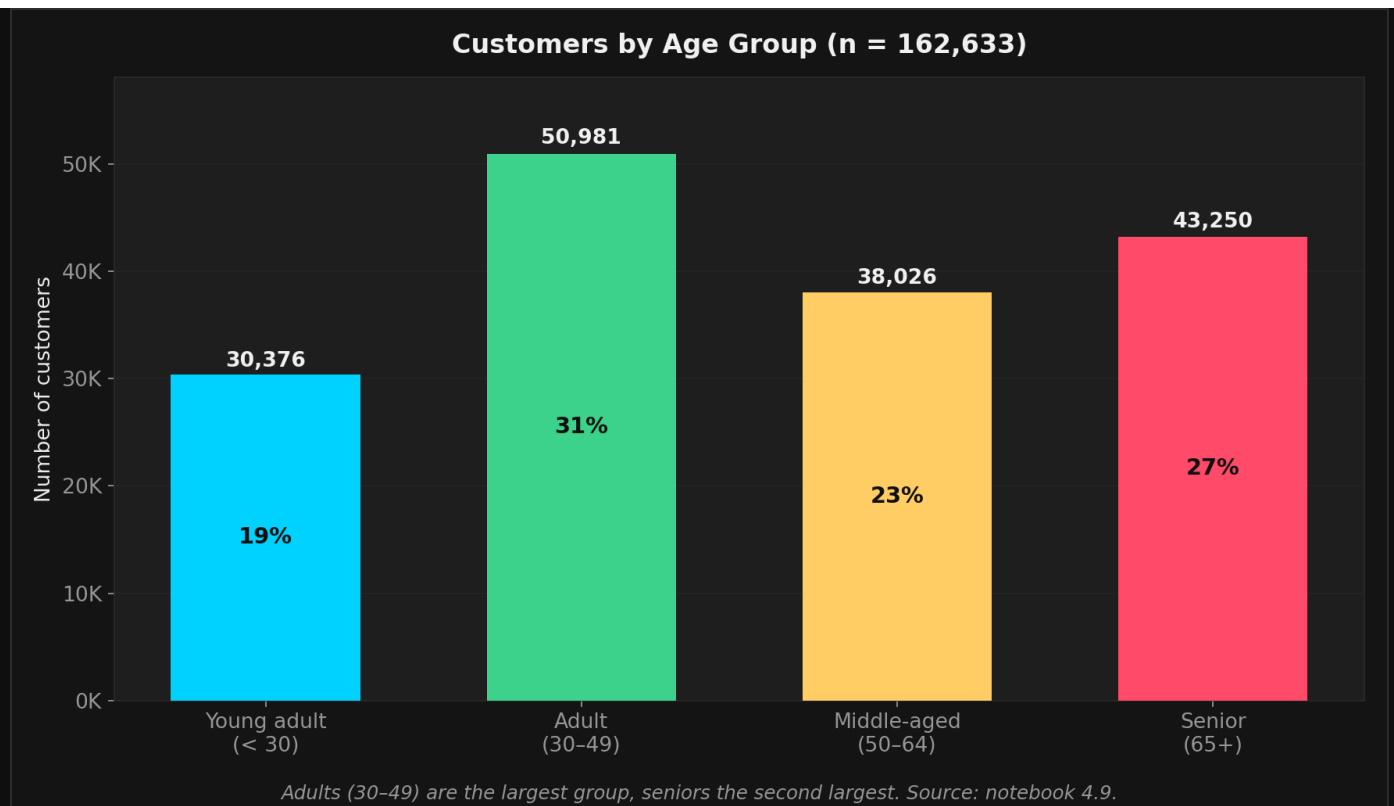


Figure 5. Adults (30-49) are the largest group (50,981 customers, 31%), seniors (65+) the second largest (43,250, 27%). Counts are customers with 5+ orders.

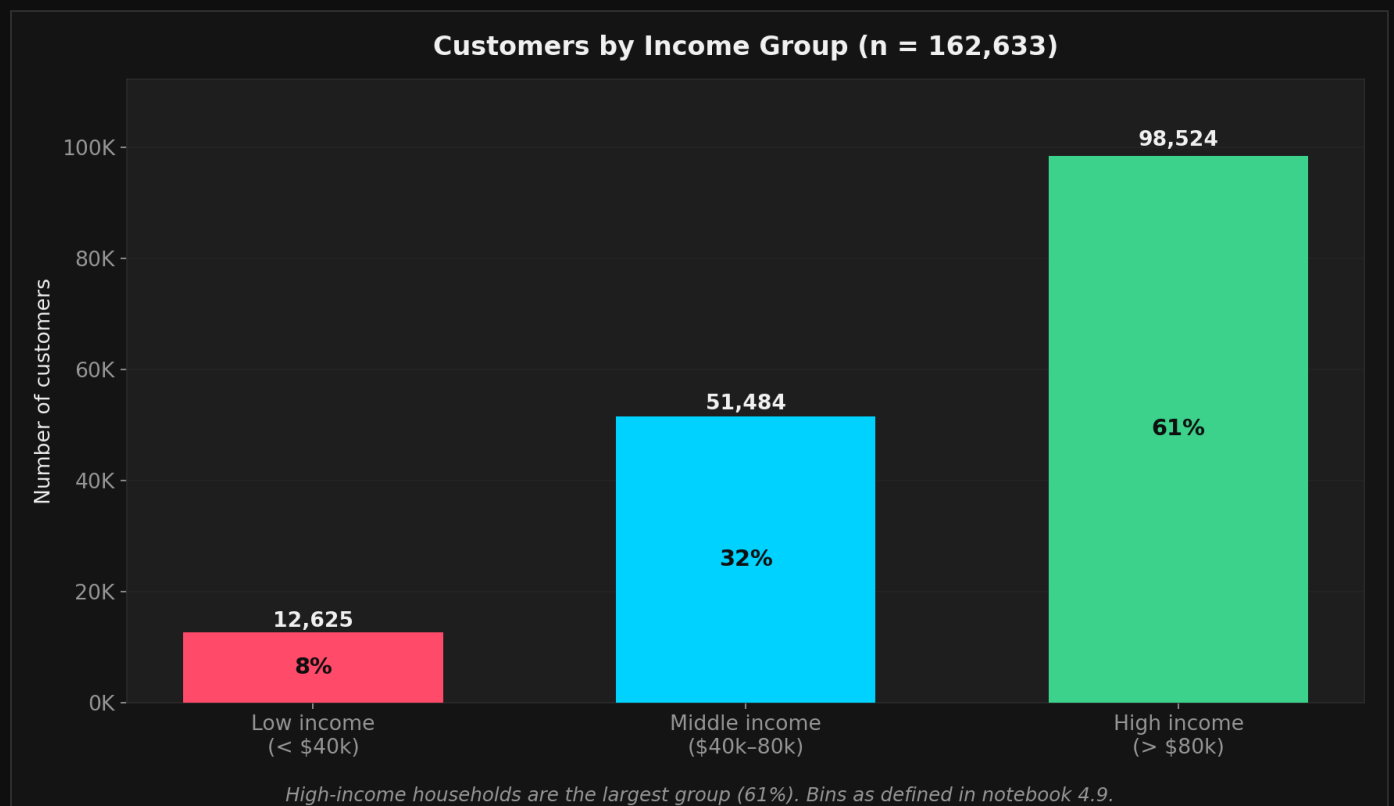
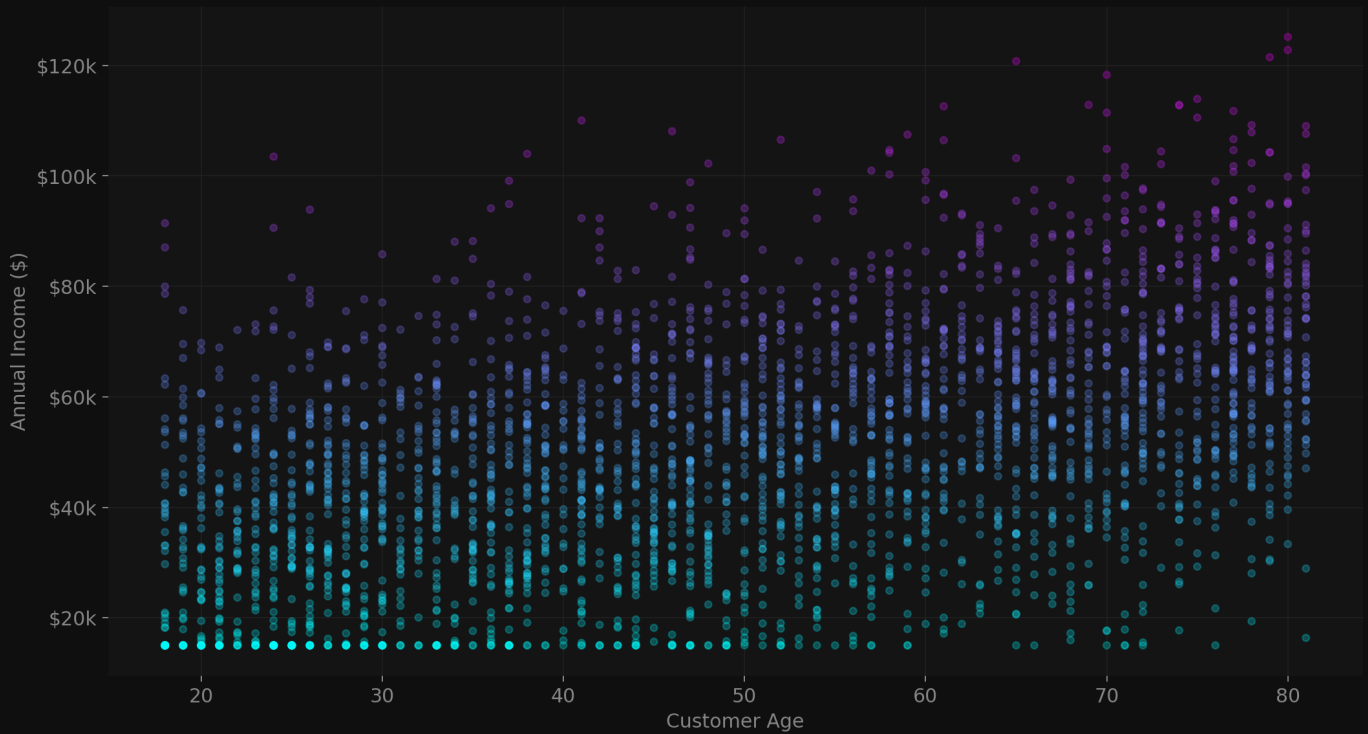


Figure 6. High-income households (> \$80k) are the largest income group by far: 98,524 customers (61%), against 51,484 middle income and 12,625 low income.

Relationship Between Customer Age and Income



Income variability increases significantly with age. No strong linear correlation — income rises but spreads widely after 30.

Figure 7. Age vs. income scatter. No strong linear correlation — income rises but becomes increasingly variable after 30.

Key Insight — Regional Segmentation

State-level analysis is too granular — 50 groups with too little data each. Mapping to 4 U.S. Census regions reduces noise while preserving meaningful geographic variation for strategy.

CUSTOMER PROFILING

Low-activity customers (fewer than 5 orders) were excluded — removing 1.44 million rows. Each remaining customer was assigned a combined profile:

```
df_filtered['customer_profile'] = (  
    df_filtered['age_group'].astype(str) + " | " +  
    df_filtered['income_group'].astype(str) + " | " +  
    df_filtered['family_status'].astype(str) + " | " +  
    df_filtered['order_frequency'].astype(str)  
)  
# Example: "Adult | Middle income | Small family | High frequency"
```

Customers by Order Frequency (customers with 5+ orders, n = 162,633)

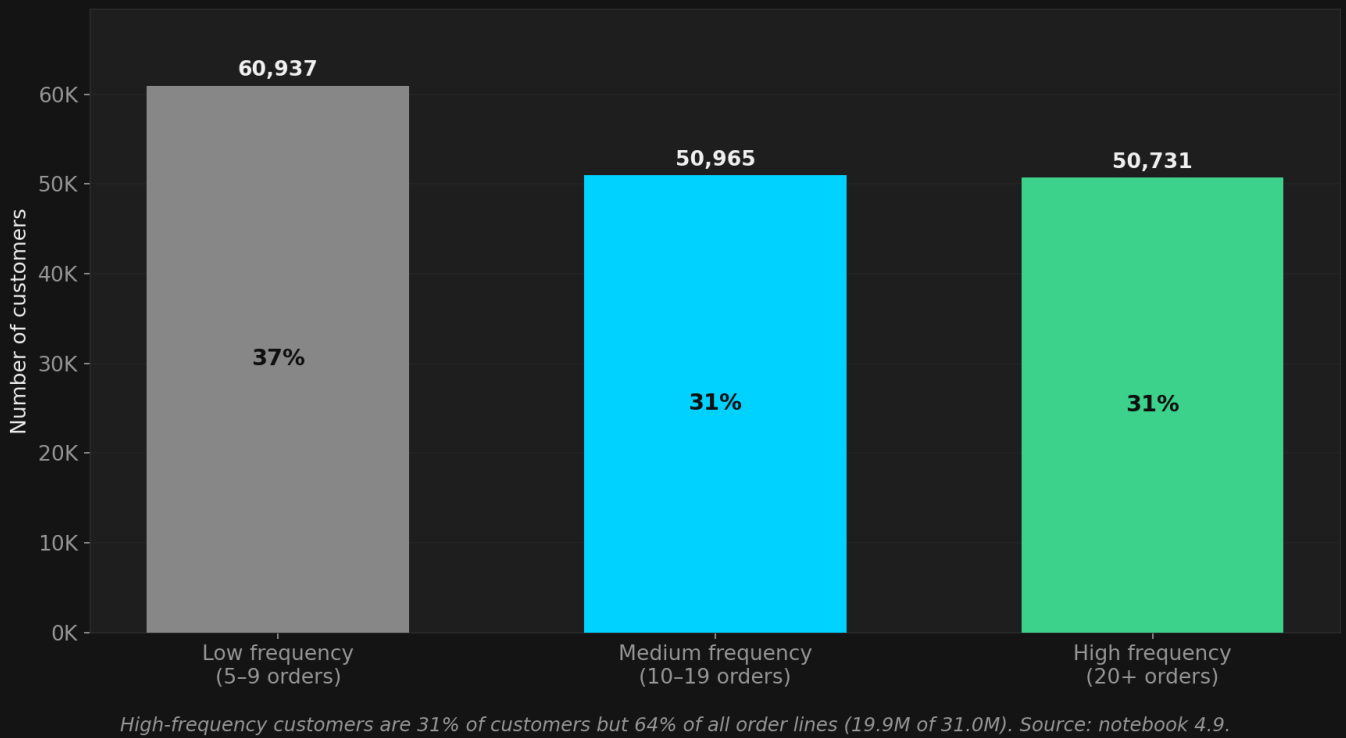


Figure 8. By customer count the three frequency tiers are similar (60,937 / 50,965 / 50,731). By volume they are not: high-frequency customers are 31% of customers but 64% of all order lines (19.9M of 31.0M).

Top 5 Customer Profiles by Order Lines (customers with 5+ orders)

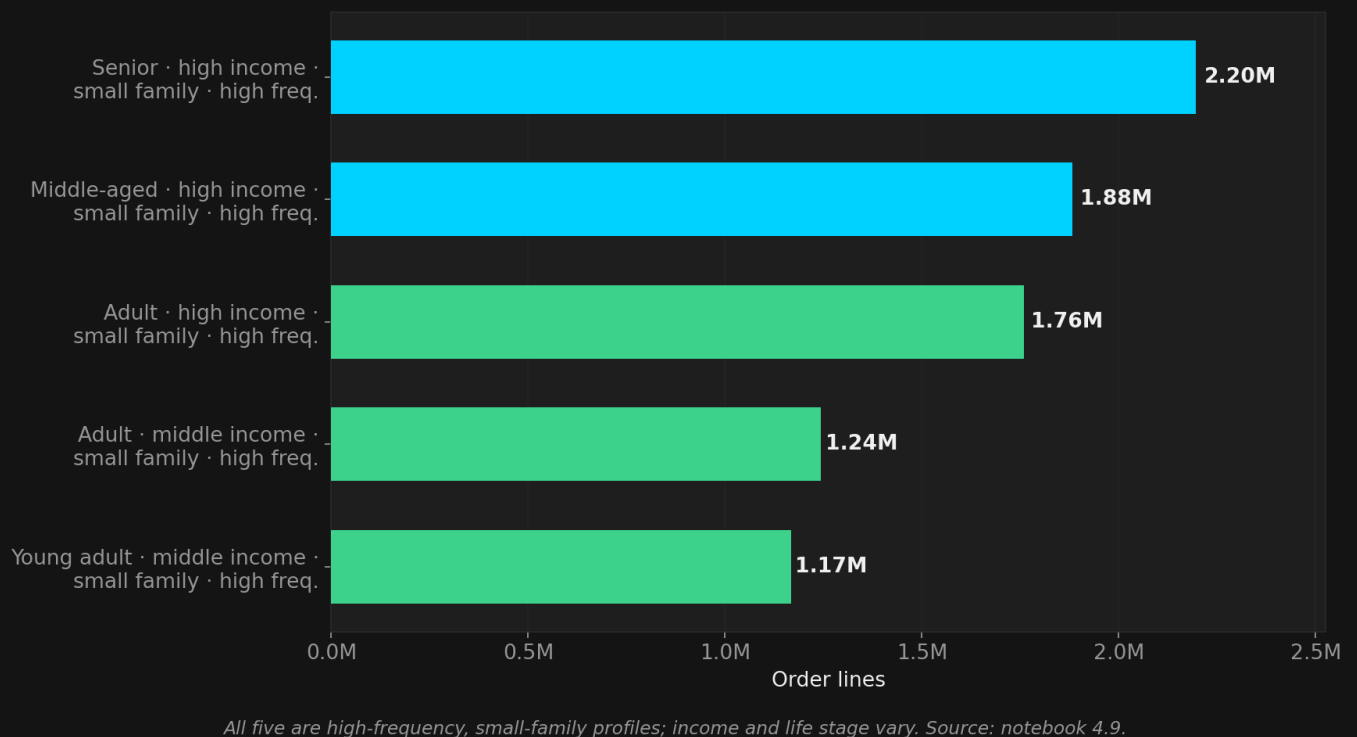


Figure 9. The five most common combined profiles. All five are high-frequency, small-family households; income and life stage vary. Behaviour, not demographics, defines the top of the base.

Key Insight — Low-Activity Filter

Excluding customers with fewer than 5 orders removed 1.44M rows (4.4% of data) but significantly improved signal quality. One-time or near-inactive users behave fundamentally differently and would skew segmentation results.

KEY FINDINGS

Finding 1 — Orders peak between 9 AM and 4 PM.

Instacart is a daytime platform. Promotions and logistics staffing should concentrate on the 9 AM–4 PM window. Peak staffing needed around 10:00–11:00 AM.

Finding 2 — Product price is stable across all hours.

Time-based pricing incentives are unlikely to change spending behavior. Promotions should target segments, not hours.

Finding 3 — New customers still account for the most order lines (37%).

Per-item spend is the same across loyalty tiers (median 7.4), so the lever is converting New to Regular and Regular to Loyal on frequency, not on basket price.

Finding 4 — Adults (30–49) are the largest age group, seniors are second.

Seniors are 27% of active customers and lead the single most common high-frequency profile. Campaigns that assume a young, family-stage base would miss more than a quarter of the active customers.

Finding 5 — High-income households are the core demographic (61%).

Income is not the constraint for most of the base. Premium placement has a large addressable audience; price-led promotions have a small one (8% low income).

Finding 6 — High-frequency customers drive 64% of order lines.

A third of the customers generate two thirds of the volume. Moving the medium tier (10–19 orders, 50,965 customers) up one band is the highest-leverage retention test.

RESOURCES

</> GitHub Repository